TECHNICAL SOLUTION BRIEF

COOLPATH

Autonomous Hyperlocal Thermal Intelligence & Worker Safety System

FORTYGUARD NVIDIA

> 120°F	< 10 ms	6 Agents	OSHA 4-Factor
STREET-LEVEL PEAK HEAT	EDGE DECISION LATENCY	AUTONOMOUS PIPELINE	THERMAL RISK MODEL

The Problem

REAL-TIME EXPOSURE VS. INERTIAL DATA

Every summer, delivery riders in cities like Phoenix work outside for hours in street-level heat that can exceed 120°F — far hotter than what a car's dashboard thermometer or a generic weather app reports. Heat illness among outdoor gig workers isn't caused by a lack of data; it's caused by a lack of anyone watching that data closely enough, in real time, for the one person standing in it. Existing tools show a number. Nobody acts on it until it's already too late.

Who It's For

RIDERS & DISPATCH OPERATIONS

CoolPath is built for delivery riders and the dispatchers who manage them. A dispatcher today has no way to know, street by street, which of their riders is in genuine danger right now versus which one is simply outside on a warm day. CoolPath makes that distinction automatically, and acts on it without waiting for a human to notice.

How We Used FortyGuard

HYPERLOCAL GROUND TRUTH & MULTI-AGENT INTELLIGENCE

We built CoolPath around FortyGuard's hyperlocal Temperature API as the ground truth for every autonomous decision. We used the heatmap endpoint with tcm, exceedance, and persistence analytics to measure not just how hot a location is, but how long it's stayed dangerously hot — the actual physiological driver of heat illness — and the environmental parameters endpoint for heat index, solar irradiance, and air quality. This data feeds a six-agent autonomous pipeline: a Heat Perception agent reads live conditions at a rider's exact position, a Risk Scoring agent computes an OSHA-grounded four-factor Thermal Risk Score, and — without any human clicking anything — a Decision Coordinator either dispatches a Reroute agent to find a genuinely cooler nearby path, or an Alert Automation agent to trigger a tiered safety alert up to a mandatory rest stop. An Explanation agent then generates a plain-language safety briefing grounded in the exact sensor readings behind the decision. Because live heatmap generation takes real time, we built a two-speed architecture: a background worker refreshes FortyGuard data on a cache every few minutes, while every live rider-safety decision reads only from that cache — resolving in under 10 milliseconds, so the system never makes a rider wait on a network call to find out they're in danger.

`Heatmap API`	`Persistence API`	`Environmental API`
TCM Analytics	**Exceedance & Duration**	**Irradiance & Air Quality**
Thermal canopy mapping & microclimate baselines	Accumulated heat load & physiological stress tracking	Solar radiation, wet bulb heat index & PM2.5 levels

The Six-Agent Autonomous Pipeline

REAL-TIME AUTONOMOUS COORDINATION & SAFETY AUTOMATION

01 Heat Perception

Reads live FortyGuard environmental readings at the rider's exact GPS coordinates, continuously tracking real microclimate variations.

02 Risk Scoring

Computes an OSHA-grounded four-factor Thermal Risk Score integrating ambient temperature, persistence duration, irradiance, and exertion.

03 Decision Coordinator

Autonomously evaluates threshold breach triggers without requiring human intervention or dispatcher review.

04 Reroute Agent

Identifies and calculates cooler nearby navigational paths through shaded, high-canopy corridors to lower thermal load.

05 Alert Automation

Executes tiered safety protocols, automatically escalating from hydration warnings up to mandatory rest stops.

06 Explanation Agent

Generates plain-language safety briefings directly grounded in the exact sensor and environmental telemetry.

Two-Speed High-Throughput Architecture

DECOUPLED INGESTION & SUB-10MS DECISION EXECUTION

1. Background Ingestion Worker

Asynchronously queries FortyGuard's Heatmap & Environmental API endpoints every few minutes to refresh local spatial caches across urban delivery zones.

2. Sub-10ms Real-Time Safety Engine

Live rider telemetry evaluates directly against the in-memory cache, enabling instantaneous autonomous actions without network lag or latency delays.

Measured Result

SIMULATION OUTCOMES & PROVENANCE VERIFICATION

Our OSHA-grounded Thermal Risk formula and FortyGuard-fed geographic risk cascade produced consistent, explainable, autonomous outcomes across our full six-rider Phoenix simulation: high-heat, low-canopy zones (like our Van Buren Corridor test case, 0% measured canopy, sustained readings above 46°C) correctly and automatically triggered Critical mandatory-stop alerts, while riders approaching genuinely shaded, high-canopy zones were automatically rerouted with a measurable temperature reduction before entering dangerous exposure. Every decision carries an explicit data-provenance flag distinguishing live FortyGuard readings from fallback estimates, so the system is never silently guessing. The result is a working, end-to-end demonstration that hyperlocal thermal intelligence — combined with autonomous multi-agent reasoning — can protect outdoor workers before harm occurs, not after.

CRITICAL RISK TEST CASE

Van Buren Corridor Simulation

Environment: 0% measured canopy cover, sustained street-level heat > 46°C (115°F+).
Autonomous Action: Triggered Tier-3 Critical Mandatory Rest Stop with cooling station dispatch.

PROACTIVE REROUTE TEST CASE

High-Canopy Shaded Corridor

Environment: Approaching high urban heat island zone adjacent to shaded tree canopy.
Autonomous Action: Dynamic reroute executed; verified measurable microclimate temperature drop.

Data-Provenance Integrity: Every safety trigger maintains transparent telemetry flags differentiating verified live FortyGuard sensor feeds from fallback algorithmic estimations.

✓ PROVENANCE VERIFIED